Mobile App Development

Individual Assignment: Milestone Mobile Apps

Apps that changed the mobile world

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The assignment at a glance

- **Individual** assignment (in addition to the team assignment - project)
- Each student takes on **one theme** (e.g. location awareness, cloud synergy, mobile payments...)
- Within the theme, they choose **one milestone app** that was a turning point in the evolution of mobile apps
  - from the early days (feature phones, J2ME, BlackBerry) to the most recent (AR, generative AI)
- **Deliverables**
  - Slide deck of **~10-15 slides** (PDF)
  - Oral presentation of **~15-20 minutes** + ~5 minutes of questions
- The key question you must answer:

> **Why was this app important, and what changed after it?**

Why this assignment?

- It builds on the **historical overview and evolution of mobile apps** (Lecture 2) and **current trends** (week 12)
- It develops:
  - **Searching, analyzing and synthesizing** information from reliable sources
  - **Critical thinking**: what makes an app "significant" rather than merely "popular"?
  - **Technical understanding**: which technologies (sensors, APIs, cloud, protocols) made the innovation possible
  - **Presentation skills** in front of an audience
- Taken together, in **chronological order**, the presentations will "build" a shared timeline of the history of mobile apps

How it works

1. **Theme declaration**: choose one theme from the pool (next slides)
   - *first come, first served* - one theme per student
   - if there are more students than themes, a theme may be assigned to 2 people, **with apps from different eras**
2. **App selection**: one of the suggested apps or your own proposal (**requires approval**)
3. **Approval** by the instructors (theme + app + 3-5 initial sources)
4. **Preparation** of the slide deck following the structure below
5. **Submission** of the PDF before the presentation
6. **Presentation** in class - presentations are scheduled **in chronological order of the app's release** (from oldest to most recent)

Theme pool (1/3)

#	Theme	What we examine	Suggested apps
1	**Location awareness**	GPS, maps, navigation, crowdsourced data	Google Maps for Mobile, Waze
2	**Context awareness & proactive assistance**	The app "knows" where you are, what you're doing, what you'll need	Google Now, Foursquare
3	**Cloud synergy & sync**	Data that "follows" the user across all devices	Dropbox, iCloud, Evernote
4	**Always-on communication & messaging**	Push, instant messaging, replacing SMS	BlackBerry (push email / BBM), WhatsApp
5	**Camera-first & visual social**	The camera as the main interface, filters, ephemeral content, stories	Instagram, Snapchat
6	**Media streaming & algorithmic feeds**	Streaming instead of ownership, algorithmic personalization	Spotify, TikTok

Theme pool (2/3)

#	Theme	What we examine	Suggested apps
7	**On-demand & sharing economy**	The phone as a "remote control" for the physical world	Uber, Airbnb
8	**Mobile payments & fintech**	Payments, banking services, NFC, tokenization	M-Pesa, Apple Pay
9	**Mobile gaming & monetization models**	Touch gameplay, premium → freemium, in-app purchases	Snake (Nokia), Angry Birds, Candy Crush Saga
10	**Augmented reality & sensor fusion**	Camera + GPS + gyroscope, AR frameworks	Pokémon GO, IKEA Place
11	**Health, fitness & wearables**	Sensors, quantified self, health platforms	Strava, Apple Health / HealthKit
12	**Privacy, security & end-to-end encryption**	E2EE, privacy by design, metadata	Signal, Exposure Notification apps (COVID-19)

Theme pool (3/3)

#	Theme	What we examine	Suggested apps
13	**Conversational & generative AI**	Voice assistants, natural language, LLMs on mobile	Siri, ChatGPT app
14	**App ecosystems, distribution & super apps**	App stores, revenue models, "apps within apps"	App Store (2008), WeChat
15	**Accessibility & inclusive design**	Screen readers, computer vision for people with disabilities, crowd assistance	Be My Eyes, Seeing AI, VoiceOver
16	**Constrained connectivity & emerging markets**	2G networks, low-cost devices, offline-first, data saving	Opera Mini, Facebook Lite
17	**IoT & device synergy**	The phone as a hub for the smart home and connected devices	Philips Hue, Nest
18	**Engineering & cross-platform breakthroughs**	Native vs HTML5 vs cross-platform, as in our lectures	Facebook app (HTML5 → native → React Native), Flutter apps

“ You may propose **another app** within a theme or a **new theme**, subject to the instructors' approval. ”

App selection rules

- The app must be a **true milestone**: not just popular, but something that changed things
- It can be:
  - an app that is **still alive** (e.g. WhatsApp)
  - an app that has **"died"** but left a legacy (e.g. Google Now, BBM)
  - an app that was the **first** at something, even if others surpassed it later
- You must connect the app to **your theme**: what did it contribute to location awareness / cloud synergy / ...
- Don't pick an app just because you use it every day. Pick an app whose significance you can **substantiate**

Significance criteria

You must substantiate **at least 3** of the following criteria, **with evidence** (numbers, dates, sources):

Criterion	Guiding question
Breakthrough for its time	What did it do first? What was considered impossible or impractical before?
Wide adoption / scale	How many users, downloads, how fast?
Technological contribution	New use of sensors, APIs, protocols, architecture, open source?
Industry impact	Did it create a category? Was it copied? Did it "kill" an industry?
UX / interaction innovation	Did it introduce a new interaction pattern (swipe, stories, cards...)?
Business model	Freemium, subscription, marketplace, ads, transaction fees?
Social / economic impact	Did it change behaviors, labor markets, access to services?
Longevity & legacy	Does it still exist? If not, what of it "lives on" in other apps?

Ethical, regulatory and critical perspective

Beyond the positives, **every presentation must include a critical assessment**:

- **Privacy** and personal data issues (permissions, tracking, GDPR)
- **Security** issues (leaks, vulnerabilities)
- **Regulatory** conflicts (e.g. licensing, competition, Digital Markets Act)
- **Social** side effects (addiction, mental health, working conditions)
- **Failures or limitations**: what didn't work, who surpassed it and why

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> A presentation that is merely an "advertisement" for the app does **not** meet the goals of the assignment.

Presentation structure (1/2)

#	Slide	Content
1	**Title**	App, theme, release year, full name, student ID
2	**The world before**	Devices, networks (2G/3G/4G), operating systems, how the problem was solved before the app
3	**App identity**	Who, when, where, for which platform, original idea - short timeline
4	**The breakthrough**	The **one** key innovation, in one sentence + demo / screenshot / video
5-6	**Under the hood**	Technologies that made it possible: sensors, APIs, backend, cloud, protocols, native / cross-platform
7	**UX & interaction**	Interaction patterns, design choices (link to Lecture 4)
8-9	**Why it mattered**	Analysis using the **significance criteria** (≥3), with evidence and sources

Presentation structure (2/2)

#	Slide	Content
10	**Adoption & business model**	User numbers, growth, how it makes (or doesn't make) money
11	**Impact**	On the industry, competitors, society, developers
12	**Critical perspective**	Controversies, privacy, security, regulation, failures
13	**Evolution & today**	What became of it? Acquisition, transformation, shutdown, successors
14	**If we built it today**	Which stack (React Native / Flutter / native), which cloud services, auth, storage - link to the course
15	**Conclusions & sources**	3 key takeaways, one discussion question, bibliography

> The structure is a **guide**: you may merge or reorder slides, as long as all points are covered.

Presentation timing (~15-20')

Section	Slides	Time
Introduction & context	1-3	~3'
Breakthrough & technology	4-7	~5-6'
Significance, adoption, impact	8-11	~5-6'
Critique, today, "if we built it today"	12-14	~3-4'
Conclusions	15	~1'
Questions	-	~5'

- **Rehearse with a timer**. If you go over 20', cut content, not your speaking pace

Deliverables & format

- **Slide deck in PDF**, ~10-15 slides (not counting the title and sources)
  - File name: `StudentID_Surname_Theme_App.pdf`
  - Any tool (PowerPoint, Google Slides, Keynote, **Marp**)
  - *Optional*: use the course Marp theme (see the repository's `README.md`)
- **Visuals**: screenshots, photos of devices from the era, architecture diagrams, short video (≤1')
  - Always with **source attribution**
- **Sources**: at least **5 reliable sources** (official blogs / announcements, press articles, scientific papers, books)
  - Wikipedia is a starting point, not a final source
- Little text per slide: slides **support** your talk, they don't replace it

Assessment

Criterion	Weight
Substantiating significance - criteria, evidence, sources	25%
Technical understanding - how it worked, which technologies, link to the course	20%
Historical context & impact - before / after, comparison with competitors	15%
Critical perspective - ethics, privacy, limitations, failures	10%
Slide quality - structure, visuals, readability	10%
Presentation - clarity, timing, audience engagement	10%
Answers to questions	10%

- **Weight in the final grade:** *bonus of up to +1 or +2 points (depending on results)*

Use of sources and AI tools

- Using AI tools is allowed for **searching, organizing ideas and language editing**
  - State on the sources slide **which tools** you used and **for what purpose**
- **Every number, date and claim** must be verified against a real source
  - AI tools often "make up" statistics and references
- You must be able to **explain orally** anything on your slides
- Plagiarism = 0 (as with all course assignments)

Timeline

Milestone	Deadline
Theme & app declaration (eclass / opencourses)	October 13
Approval + presentation date assignment	October 20
PDF submission	**48 hours** before your presentation
Presentations (in chronological order of the apps)	November 3 - December 15

- In your declaration include: **theme, app, release year, 3-5 initial sources**

Tips & common mistakes

✓ Do

- Start with the **"why"**, not the "what"
- Show the app **as it was back then** (early versions, old screenshots, launch keynotes)
- Use **numbers with sources**
- Connect to what we do in the course (native vs cross-platform, auth, cloud, UX)
- Close with a **discussion question**

✗ Avoid

- A "user guide" for the app
- Walls of text on slides
- Numbers without a source or date
- Only positives, without critique
- Running over time
- Reading off the slides

Questions?

Happy researching!

